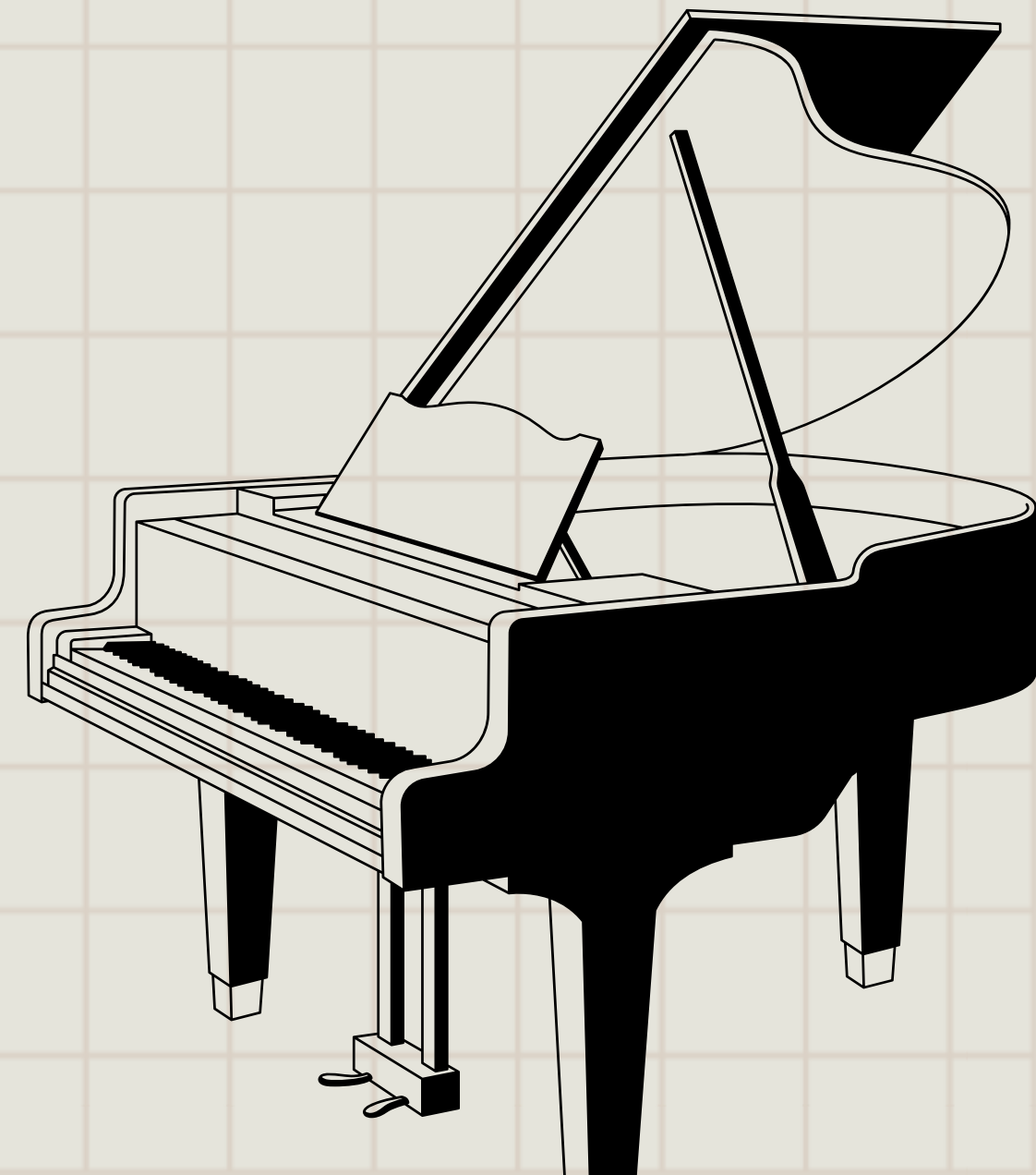


MELODY-TO-PIANO

TRANSFORMER-BASED FULL
PIANO ACCOMPANIMENT
GENERATION FROM SIMPLE
MELODY

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PROBLEMS



Can attention-based sequence models learn accompaniment structure directly from symbolic datasets?



Does explicit harmonic conditioning improve generation quality?



Earlier studies on neural symbolic generation had limitations in capturing long-range structures and hierarchical repetition patterns.



Music Transformer introduced relative positional attention, allowing more effective long symbolic music sequences modeling by capturing relative timing relationships.

TASKS

- Develops a conditional sequence-to-sequence learning framework that produces automatic piano accompaniment generation.
- Requires the model to decode an event-based MIDI token sequence that demonstrates high harmonic compatibility and rhythmic synergy with the source melody.
- Focuses on melody-conditioned accompaniment prediction, utilizing chord conditioning as an explicit structural guide.

Melody token sequence



Structured, polyphonic piano accompaniments sequence

The **POP909** dataset provides:

- Aligned melody,
- Accompaniment,
- Beat, and
- Chord annotations.

001/
001.mid
beat_audio.txt
beat_midi.txt
chord_audio.txt
chord_midi.txt
key_audio.txt
versions/
001-v1.mid

DATA



MIDI DATA EXAMPLE

The image displays a digital piano learning application interface. At the top, a control bar includes buttons for play/pause, a 10-second rewind, a volume slider set to 100% (90.0 bpm), a pencil icon for editing, a progress bar from 0:01 to 3:13, a speaker icon, a settings gear, and a full-screen button. Below the controls is a three-staff MIDI editor. The top staff is labeled '4', the middle '3', and the bottom '2'. The bottom staff also shows '2/4' time signature and 'C major 90 bpm'. The MIDI data is represented by red and blue vertical bars of varying heights and widths, indicating note duration and pitch. At the bottom, a digital piano keyboard is shown with white and black keys, each labeled with a letter (A through C). Below the keyboard, a dark blue banner contains the text 'Drop your MIDI file and start learning instantly'.

Drop your MIDI file and start learning instantly

APPROACH

Task Formulation

Let:

- $M = (m_1, \dots, m_T)$ denote melody token sequence
- $P = (p_1, \dots, p_S)$ denote accompaniment token sequence

Conditional distribution model:

$$P_{\theta}(P|M)$$

where P_{θ} is a Transformer encoder-decoder model implemented and trained using token-level cross-entropy loss with teacher forcing.

Symbolic Representation

MIDI tracks are converted into event-based symbolic tokens:

- Note onset,
- Note offset,
- Temporal shift, and
- Velocity events.

A segmentation step divides songs into fixed-length segments of 4–8 bars.

Tracks used for main task formulation:

- **MELODY**: input sequence
- **PIANO**: accompaniment target

The BRIDGE track is excluded as it serves as an auxiliary melodic layer.

Model Variants

After symbolic conversion, songs are split into training (80%), validation (10%), and test (10%) partitions.

1. Rule-Based Baseline

Creates simple accompaniment patterns from chord labels.

2. Melody-Conditioned Transformer

Uses Transformer encoder-decoder, where the encoder processes melody tokens and the decoder predicts accompaniment tokens autoregressively.

3. Chord-Conditioned Transformer

This variant also includes chord tokens in the encoder input.





EVALUATION METRICS

Note Precision, Recall, F1

Let:

- N_{pred} denote the set of predicted notes
- N_{ref} denote the set of reference notes

$$P = \frac{|N_{pred} \cap N_{ref}|}{|N_{pred}|} \quad R = \frac{|N_{pred} \cap N_{ref}|}{|N_{ref}|} \quad F_1 = \frac{2PR}{P+R}$$

Onset Precision, Recall, F1

The same formulation applies only to onset events. These metrics ignore note duration and instead focus directly on rhythmic placement.

Chord-Tone Ratio

Measures harmonic consistency.

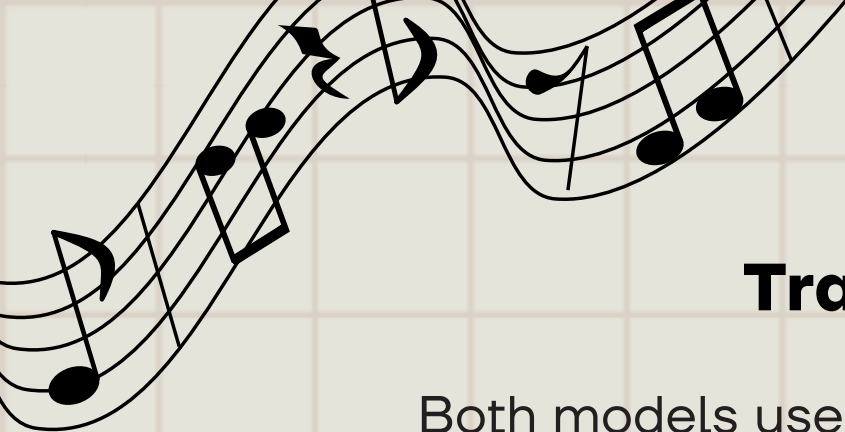
Let:

- G denote the set of generated notes
- $\mathcal{C}(t_g)$ denote the set of pitch classes of the chord active at the onset time of note g

$$CTR = \frac{1}{|G|} \sum_{g \in G} \mathbf{1}(\text{pitch}(g) \in \mathcal{C}(t_g))$$

Register Balance Error

Absolute deviation between predicted and reference note proportions in lower and higher pitch ranges.



Training Performance

Both models use **Adam optimizer** with **early stopping** based on validation loss.

Performance Comparison:

Model Variant	Best Validation Loss	Convergence
Melody-Conditioned	1.0854	Rapid
Chord-Conditioned	1.0770	Slightly slower

RESULTS

Evaluation Metrics

Metric	Melody-Conditioned	Chord-Conditioned
Note Precision	0.0689	0.0698
Note Recall	0.0897	0.0849
Note F1	0.0759	0.0748
Onset Precision	0.1780	0.1808
Onset Recall	0.2423	0.2305
Onset F1	0.1995	0.1976
Chord-tone ratio	0.6684	0.6946
LH ratio error	0.1242	0.1192
RH ratio error	0.1242	0.1192

BASELINE MIDI EXAMPLE

The image displays a MIDI software interface with a piano roll and a keyboard. The piano roll has three staves labeled 1, 2, and 3 on the left. The top staff (3) contains several green notes. The middle staff (2) contains a dense cluster of green notes. The bottom staff (1) contains a few green notes. The keyboard at the bottom is labeled with letters A through C. The notes are green and have a duration of approximately 0.5 seconds. The interface includes a top bar with various controls: a hand icon, a undo/redo icon, a zoom in/out icon, a tempo display (100% 120 bpm), a play/pause icon, a volume icon, a settings icon, and a full screen icon. The tempo is set to 120 bpm. The time display shows 0:00. The keyboard is labeled with letters A through C, and the notes are green.

3

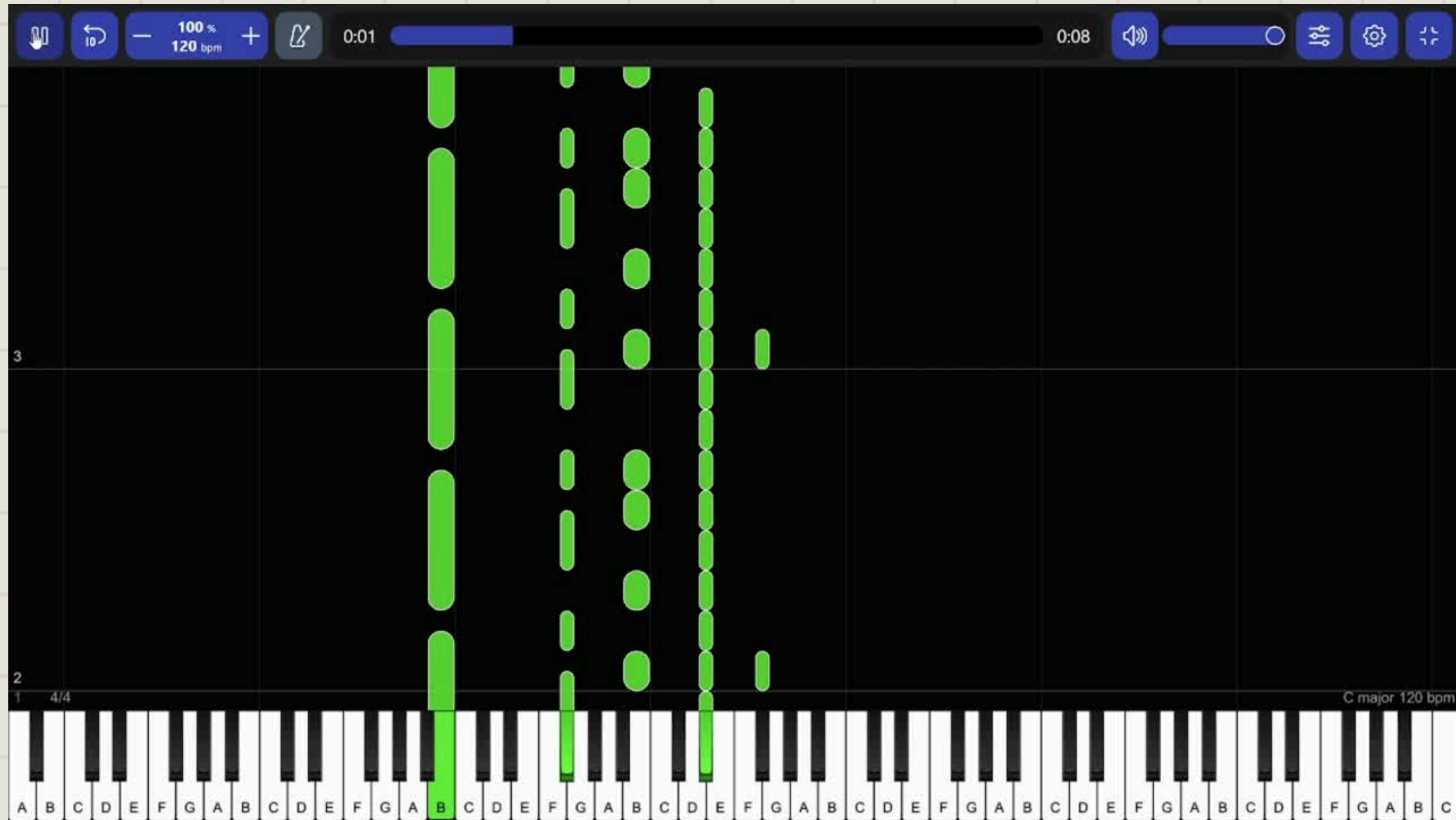
2

1 4/4

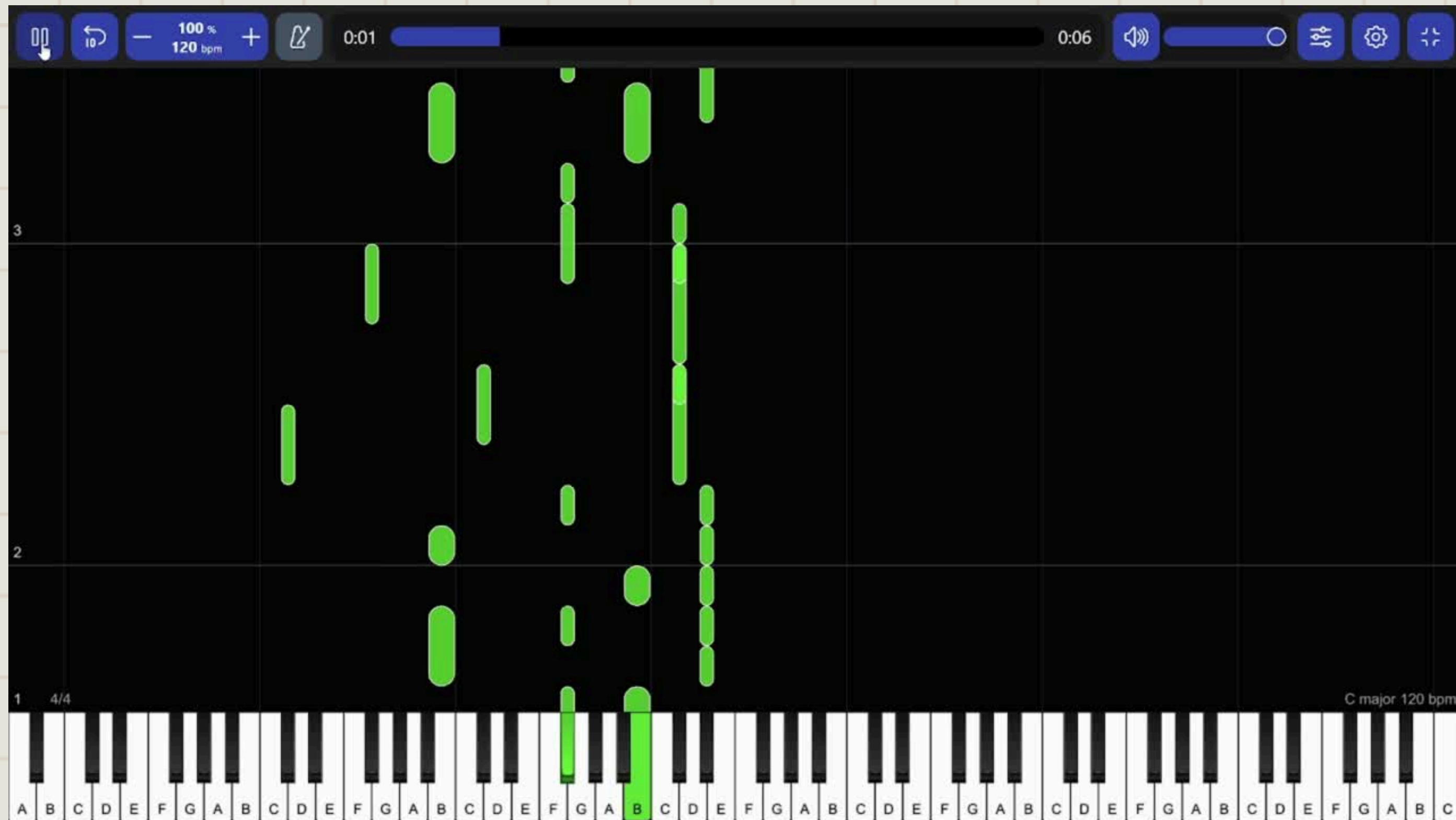
C major 120 bpm

A B C D E F G A B C D E F G A B C D E F G A B C D E F G A B C D E F G A B C

MELODY-CONDITIONED MIDI EXAMPLE



CHORD-CONDITIONED MIDI EXAMPLE

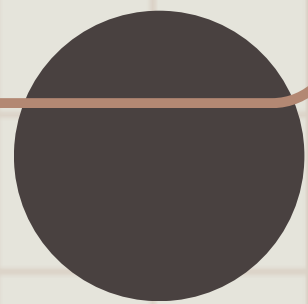


- 
- Non-Trivial Structure: Generated MIDI outputs exhibit complex accompaniment patterns rather than simplistic or random noise.
 - System Integrity: Successfully avoided common "model collapse" issues:
 - No Empty Outputs: Models consistently generate active musical content.
 - Pitch Variety: Avoids degenerate "single-pitch" repetitions.
 - Musical Realism: Register allocation stays within realistic, playable piano ranges.
 - Pipeline Validation: Confirms the end-to-end training and inference workflow is functioning correctly for structured symbolic music.

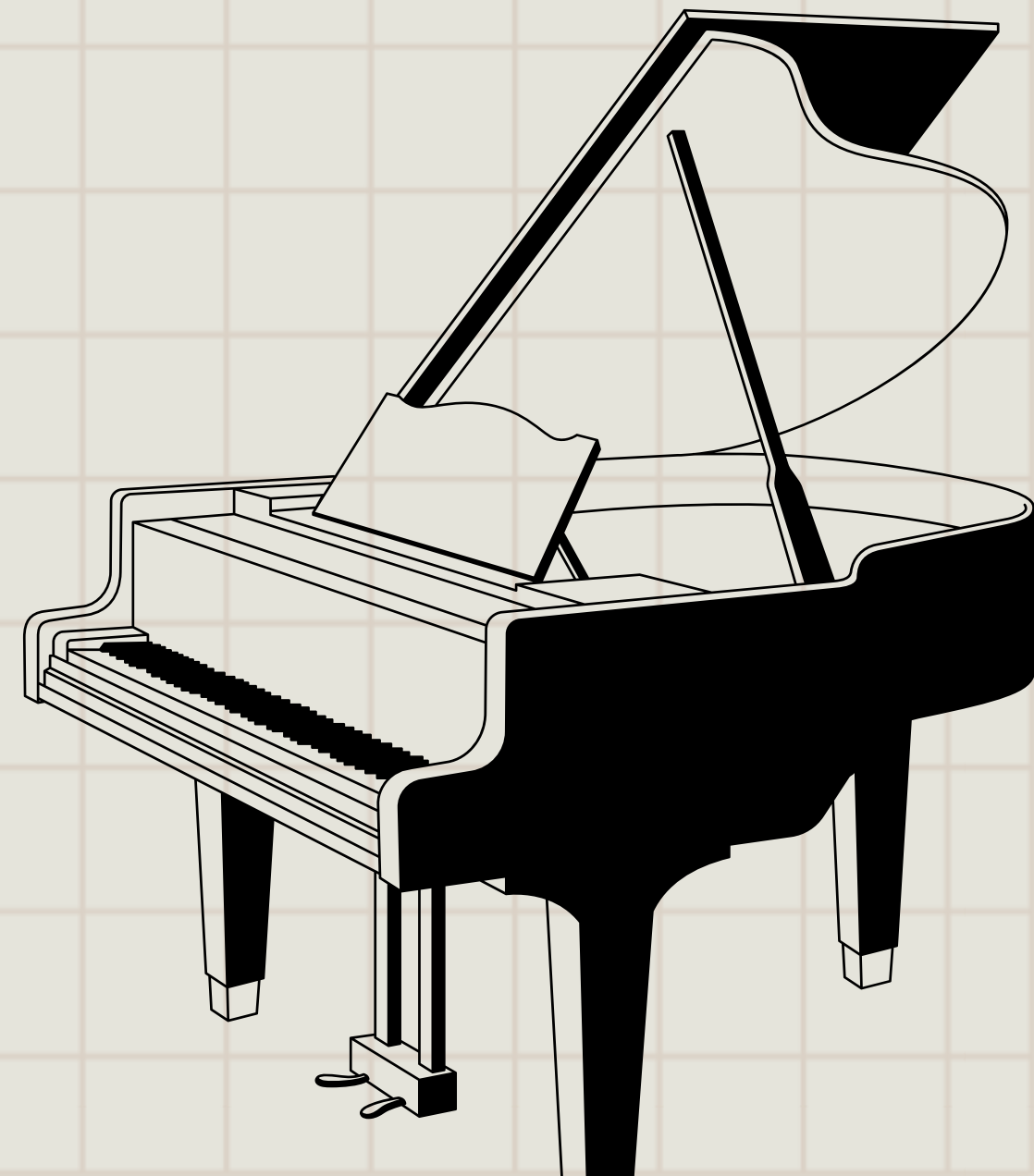
ANALYSIS

CONCLUSION

Overall, the experiments demonstrate that melody-conditioned Transformer models can learn meaningful accompaniment structure from symbolic datasets, and that harmonic conditioning improves chord-level compatibility. However, precise polyphonic reconstruction remains challenging, as reflected in the gap between onset-level and note-level metrics.



THANK YOU



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